

TED Lecture: Albert-László Barabási

Text

[Music] so in many ways a broken car is not so different from a disease when the engine is smoking and the lights don't come up there's a fundamental difference however between humans and cars if I can get my car to a mechanic I can be pretty certain that they can fix it which is much more than we can say about many of our diseases today so why can a mechanic with much less education and much less box than a doctor fix our C while our doctor often let us go with diseases persisting in our body well there are a number of things that actually the mechanic has that the doctors doesn't have right now first of all it got a parts list it has a blueprint telling us how the pieces connect together it has Diagnostics tools to figure out where the comp you know what which is broken and what is healthy and it has means essentially to replace the parts now let's think about it what which of these components are available to our doctor today well the good news is that they finally got the parts list that was the outcome of the Human Genome Project and when the human genome was actually mapped about 10 years ago we thought it's going to be easy from now from the parts we will have essentially the W Bonanza that we need to fix us humans but of course the reality is sinking we also real thought that these many pieces will eventually give us lots of drugs in 2001 or 20 000 the year before the Genome Project was unveiled the FDA approved about a 100 drugs per year we thought this number could only go up it could only just increase yet the reality is in kin the number of approved drugs essentially in the last 10 years went from 100 before the genome to about 20 per year in hindsight the reason is pretty clear it's not enough to have the parts list we also need to actually have the how the pieces fit together that is we should not look at this picture but rather we should be looking at how the wiring diagram of the car looks like how the wir diagram of the of our cells actually look like how the genes and the proteins and the metabolites link to each other forming a consistent Network because this network which I'm going to try to tell you today is really too key to understanding

Text
human diseases now the problem is that if you look at this map you soon realize that you
know it looks completely random randomness certainly has the upper hand but down the line
it is not I believe that there's a deep order behind behind this wiring diagram and
understanding that order is the key to understand human diseases now I'm a physicist and
the conventional wisdom is that a physicist I should be studying very big objects starts
quazars or very tiny ones like the Hig Bon and quar yet about a decade ago my interest
has turned to a completely different subject complex systems and networks and that's
because our very existence depends on the successful functioning of systems and networks
behind us and I also believe that the scientific challenges behind complex systems and
networks are just as DS as behind quarks or quasars so I started looking at the structure
of the internet telling us how many many uh computers are linked together by uh various
cables I looked at the structure of the social network telling us how the society is
wired together through many friendship and other linkages and eventually I started
looking at the structure of the cell telling us you know how our genes and proteins link
to each other into a coherent Network and through that path I arrived to human diseases a
path that is really taken by physic IST now the fundamental question that really comes up
from that is you know how do we think about diseases in the context of these very very
complicated Network and from that let me turn to a map that we all understand probably
the most famous map out there which is the map of Manhattan now in many ways Manhattan is
rather different from a uh from a cell but let's for a moment carry with me and let's
believe together that this is really not a map of myam but it's a map of the cell where
the intersection show as notes are the genes and the proteins and the street segments
that connect them corresponds to the interactions between them now down the line you know
this is not so different from what happens in our cells the busy life of Manhattan very
easily Maps into the crowded life of the cell where molecules interact with each other
and recombine and transport and so on so there's lots of similarities on the surface
between them and if we look in Manhattan we also realize that action is not uniformly
spread within the city if you want to go for example a theater you don't go to any parts
of Manhattan you would go to the Theater District because that's where most of the
theaters are that's where the shows are if you want to buy an artwork you will not be

Text
actually going anywhere within the city but you would be going to the gallery District
because there's one small region of the town that that has most of the highend Galleries
and that's where most artwork is for sale the same is true in the cell you know its
functions are not spread uniformly within the network but there are other Pockets within
the network that are responsible for particular functions and their breakdown potential
release to disease so the way you think about disease in the context of the network is to
think that there are different regions that correspond to different diseases of this map
so for example you could say that cancer stay somewhere around Wall Street and and
bipolar disease would be somewhere around Time Square and you know if you think of asthma
a respiratory disease it would be somewhere up next to the Washington Bridge you know
where really Manhattan breeds his people and cars into New Jersey and the Bronx now under
normal conditions Manhattan is full of traffic and you know that's how the cell looks
like our normally but if we have defects some genes breaking down that corresponds to
some of the intersections not working and soon enough we will get a very typical
Manhattan disease a traffic jam this is again not so different from what happens in our
cells because there are many different ways you can actually get the same phenotype in
the same ways there are many different ways you can actually get a disease for example
there was a famous study by Bert Wen Stein's Group which which uh sequenc about 300
individuals who all had coloror cancer they had the same phenotype so therefore the
expectation was that all of them would have probably the same mutations in the same genes
yet the study showed that not only they didn't have the same set of mutations but the
mutations were all in different groups of genes there were no two individuals who would
be actually have the same genes exactly the same group of genes uh uh defected the only
way to understand how is it possible that many different genes broken down in different
combinations linked to the same disease is to think think in terms of Manhattan is to
think in terms of disease module and to really have the wiring diagram of the disease
module to understand you know the breakdown modes of the particular system now if we
really believe that particular uh picture the next step for us is really to say how do we
proceed from here and it's very easy get a map find the disease module and Drug it now of
course you do realize there's a catch here and the catch of course is that unlike for

Text
Manhattan we don't have yet a map for the uh for the cells I mean we do but some of the maps we have are very incomplete for example the best protein interaction map we have right now believe we believe that has only 5% of the links that are supposed to be in our cells now having part five 5% of the links means that we're missing n 95% of the links and that has dramatic consequences on the system let me illustrate that on Manhattan let's go ahead and take 95% of Street segments and remove it from the map let's see what does it do to Manhattan and the consequence is obvious the network will be broken into tiny pieces and as a result the modules the Wall Street neighborhood and timec neighborhood that were clearly distinguishable before will be all over the map you you don't know anymore where your disease module is so what can we do then well first and foremost we must improve on our maps and that's what my colleague uh Mark Vidal does at Den Farber Cancer Institute who in the last 20 years has developed a whole series of automatic tools to systematically map the protein interactions within the cell one of the very important component of the cellular network as a result of his work in about a few years ago we got what we call the 5% map the one I referred to earlier this year he's about to unveil another landmark the 20% map of the human cell and if we let him to stay in truck actually he will do the fall Network he may take a decade or two to get to it but eventually you know with the effort of he and many others we will get a map but what until then should we just wait for for him to finish the work and the answer is well not really there's lots of things we can actually do using the existing Maps so this is how the map looks like right now this is all the interactions that we leave we believe should be actually in the cell and in order to understand where diseas is lying that what I'm going to do next is that I will go ahead and place on this map a particular disease in this case Asma Asma is a respiratory disease that leads in a coughing shortness of breath and many other symptoms and there has been a tremendous amount of research understanding the genetic origins of masma so therefore we know about 100 genes that are known to be associated with Asma so if we put them on the map and I'm showing them now here as purple notes then we would expect them they would be all together because they really should correspond to our disease module but they're not they're all over the map and the reason why there W the map is pretty clear because we are missing 95% of the interactions so the

Text
critical links that would really hold them together in one module are all gone they're
not there yet so what is that we can do next well we can use the power of the network
that they are really built into the network and try to figure out other genes who may
also be involved in ASA who about whom we don't know yet and that's exactly what we did
next we took this map and we run algorithm through that that really extract the
information from this map and identify what you see in front of your eyes the asma module
within the cell now if we know the Asma module from there we can uh understand the D
disease mechanism the disease Pathways and one day can actually help us understand the
drugs but this is not only true for Asma not only Asma is located well in within the
network you can take some other diseases for example COPD and try to do the same thing
COPD is often called a smoker's disease because smokers are at very high chance of
getting it and has somewhat similar symptoms to asthma so not surprisingly the net it
seems to be that the two modules are significantly overlapping and they're certainly in
the same region of the network we do expect however to have other diseases that would
been a completely different part of the network and what is crucial here is to understand
that the relationship between these diseases to what degree they overlap and how they
relate to each other is really crucial to understand you know what is you know what how
how they relate to each other and whe whether they are similar or very different from
each other so one way to look at it is to you know let look at the relationship between
all diseases and that's what I'm showing you here here in the next slide every node
corresponds to a uh to a particular disease and two diseases are connected to each other
if they share a gene why would you do that well because if they share a gene then very
likely their disease module overlaps and therefore they must be in the same region of the
network and what is amazing about this map is that you see that there are links between
completely apparently unrelated diseases which is telling us that if you really want to
kind of treat two you have two diseases and you want to treat them today you may go to
different doctors different hospitals different floors but down at the level of the cell
they are not independent from each other they're connected because they're rooted in
someh out in the same neighborhood so what this is telling us this disease map how we
call it is that if we want to understand disease we should not be looking really at the

Text

you know what we normally look at but we should be really looking at the network within our cells this is the one that really matters this is the one that really should tell us how to classify diseases you know we probably got it fundamentally wrong it's not heart it's not Brains it's not kidneys you know sooner or later we must abandon this organ based description of the disease and turn to what really matter we should stop training cardiologist and and uh and neurologist and the rather the doctor of future really needs to become a bit of network list to understand where diseases are lying within that Network and how they relate to each other so I personally believe that we need a new medicine to truly execute the Paradigm change that genomics allowed us to really achieve I would call it Network medicine and I think it's really within the our footstep to really kind of do and Achieve that I also think that Network medicine will not only help us understand the mechanism disease but it will affect all aspects of hair care you know from the role of the environment all the way to how we actually deliver care to a particular patient so climbing back to our original question you know the good news is that doctors are increasingly letting many many of the tools that the car mechanics has today you know if you think about it the genomics provides the parts list metabolomics and proteomics provides diagnostic tools and gene therapy is really giving us the way to one day to replace the components with with the pieces that are not broken but a car mechanic would be useless without the blueprint and in the same way I believe that to truly understand diseases we need to give in the hands of our doctors the map now I'm a physicist and a network scientist I'm not a medical doctor hence I will never cure any of your diseases I can help however decipher the map the real Book of Life the book that is currently missing most of its pages but once we learn to read it we'll get much closer to the secret of life and curing disease thank you very [Applause] much